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USE OF AN OPTICAL PROBE FOR TIME-RESOLVED *IN SITU* MEASUREMENT OF LOCAL AIR-TO-FUEL RATIO AND EXTENT OF FUEL MIXING WITH APPLICATIONS TO LOW NO_x EMISSIONS IN PREMIXED GAS TURBINES

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The lower temperatures associated with lean premixed combustion generally lead to lower NO_x emissions; however, the benefit of lean premixed combustion may be lost if the fuel and air are poorly mixed. In this paper, we describe the development of an inexpensive fiber optic probe capable of measuring the extent of mixing. The fuel concentration is determined by laser light absorption at 3.39 μm over a short path length created by using infrared transmitting fiber optics.

A hydrogen-piloted, CH_4-in-air turbulent flame with a variable fuel injection location is used to vary the degree of mixedness at the burner exit. We use the optical probe to measure the level of mixedness (nonreacting) at the burner exit. The level of mixing and the mean concentration profiles are also measured by using planar laser–initiated Rayleigh scattering. NO_x measurements are reported for several mixing distances. We show that at lean conditions ($\bar{\phi} = 0.6$), incomplete mixing causes a dramatic increase in NO_x production because of the exponential temperature dependence of NO_x formation about $\bar{\phi} = 0.6$.

We also numerically investigate how the extent of mixing affects NO_x production at various equivalence ratios and pressures. Modeling the effect of incomplete mixing on NO_x formation is done with a distribution of ϕ convolved with numerical results from a perfectly stirred reactor in series with a plug flow reactor. The model does an excellent job of predicting the NO_x increase caused by incomplete mixing at lean conditions. Model predictions at higher pressures that are typical of gas turbine conditions show good agreement with available data. In particular, for lean premixed combustion, NO_x is *not* a function of pressure if the air and fuel are well mixed.

Introduction

Lean premixed combustion is emerging as a method of meeting increasingly stringent air-quality regulations for nitrogen oxides ($NO + NO_2 = NO_x$) emissions [1]. Since NO_x formation is a strong nonlinear function of the flame temperature, lean combustion yielding a lower flame temperature will lead to reduced NO_x emissions.

Extent of Fuel Mixing in Premixed Combustion

In poorly premixed combustion, the stream approaching the flame zone contains localized pockets that are richer or leaner than the mean of the mixture. Lyons [2] showed that in premixed flames with an overall equivalence ratio less than $\phi = 0.7$, variations in the mean concentration profile cause an increase in NO_x emissions. Using point LIF on NO_2-seeded flows, Gulati and Warren [3] measured temporal concentration fluctuations in a turbulent jet. Fric [4] used the NO_2 LIF technique to show that for *lean* premixed combustion, NO_x emissions are increased when temporal nonuniformities of the fuel-air mixture are present. Graham [5] and Hase et al. [6] have observed an opposite effect for burners operating at near stoichiometric conditions; the NO_x emissions increase as the mixing improves.

The extent of mixing is quantified by the *unmixedness U* [7],

$$U \equiv \frac{{c'_{\mathrm{rms}}}^2}{\bar{c}\cdot(1-\bar{c})} = \left(\frac{c'_{\mathrm{rms}}}{\bar{c}}\right)^2 \left(\frac{\bar{c}}{1-\bar{c}}\right) \qquad (1)$$

where $\bar{c}$ is the mean fuel concentration and c'_{rms} is the rms fluctuation of the fuel concentration. For a perfectly mixed stream, $U = 0$; for a perfectly unmixed stream, $U = 1$. The term $[\bar{c}(1-\bar{c})]^{1/2}$ is the maximum rms possible of a signal that is bounded by 0 and 1. More generally, the maximum rms of a signal is $[(\bar{c} - c_{\min})(c_{\max} - \bar{c})]^{1/2}$ [8].

The NO_x formation rate is strongly dependent on the local equivalence ratio ϕ. To show this, we model the NO_x production for CH_4-in-air premixed flames by using a 2-ms residence time perfectly stirred reactor (PSR) in series with a 4-ms residence time plug flow reactor (PFR), as was done by Steele et al. [9].

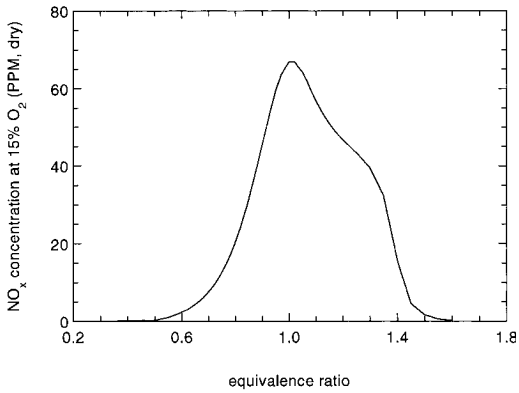

FIG. 1. NO_x at 15% O_2 versus equivalence ratio as calculated by the PSR/PFR model. The PSR has a residence time of 2 ms, and the PFR has a residence time of 4 ms. The inlet reactants are at 300 K and 1 atm.

GRI-Mech V2.11 [10], which includes nitrogen chemistry, is used for the calculations. As depicted in Fig. 1, the model predicts that the NO_x formation rate peaks at near-stoichiometric conditions.

For *lean* premixed combustion, the aforementioned LIF by Fric [4] shows that an increase in unmixedness will increase NO_x formation. Fluctuations around the mean produce regions that burn slightly richer and leaner than the overall mean equivalence ratio. The leaner regions have lower NO_x production rates than the mean equivalence ratio; however, the increase in NO_x production in the richer pockets outweighs the reduction in the leaner pockets. The net increase in NO_x emissions results from the nonlinear equivalence ratio dependence of NO_x production.

Interestingly, perfect mixing is *undesirable* when the mean stoichiometry is near $\phi = 1$, such as occurs in utility burners [5,6]. At near-stoichiometric conditions, fluctuations about the mean concentration will produce regions that are lean and rich. These lean and rich pockets will produce lower NO_x, as shown in Fig. 1. The focus of this work is to provide more detailed information on the unmixedness effect on NO_x emissions and to develop a tool to accurately measure the unmixedness.

Laser Absorption of Hydrocarbons

The use of laser absorption in combustion systems has been demonstrated by several authors including Schoenung and Hanson [11] and Nguyen et al. [12] for the measurement of CO and temperature. Several authors have studied the absorption by methane of the infrared He-Ne laser transition at 3.39 μm [13,14,15]. Emmerman et al. [16] applied absorption tomography to a methane jet into air and obtained two-dimensional maps of methane concentration. Tsuboi et al. [17] showed that ethane, propane, *n*-octane, and several other hydrocarbons also strongly absorb the 3.39-μm He-Ne laser. More recently, Drallmeier [18] used the infrared He-Ne laser in conjunction with a visible He-Ne laser to measure the fuel vapor concentration of a fuel spray. The visible laser was used to account for absorption caused by droplets in the optical path.

The Beer–Lambert law relates the transmitted intensity I to the original intensity I_o by

$$\frac{I}{I_o} = \exp(-\alpha \cdot L \cdot P \cdot x_i) \tag{2}$$

where L is the path length, α is the absorption coefficient, P is the pressure, and x_i is the mole fraction of the hydrocarbon. Because the absorption coefficient is on the order of 10 atm^{-1} cm^{-1} for most hydrocarbons [17], measurable absorption for combustible mixtures occurs with a path length of only 5 mm. The short path length defines the spatial distribution. More important, because the gas is not drawn through a sampling line, determination of the temporal characteristics is limited only by the detector and data acquisition speed. This allows measurement of the extent of mixing. Greater detail on the spectroscopy of methane in this wavelength regime can be found in the work of Cline and Varghese [19] and Perrin and Hartmann [20].

Experimental Setup

Lean Premixed Combustion System

The burner tube with annular pilot used by Karlovitz (described in the book by Lewis and von Elbe [21]) and later by Dibble and Hollenbach [22], shown in Fig. 2, has several important characteristics. The annular pilot stabilizes the flame at the same location at various equivalence ratios and levels of mixedness. Second, the combustor allows optical access for the planar Rayleigh scattering that is described in a later section. We have modified the Karlovitz burner by injecting the fuel closer to the burner exit. By changing the mixing distance l, the rms of fuel concentration (i.e., the extent of mixing) can be changed without changing the radial mean concentration profile.

As shown in Fig. 2, the burner has a methane injection tube with $d = 3$ mm, an air tube with $D = 27$ mm, and a hydrogen pilot with annular thickness of $\delta = 3$ mm. The upstream mixing distance l is varied between 33 and 63.5 cm. A fully premixed flame was produced by mixing the methane with the air well upstream and not using the methane injection tube. The methane tube bulk velocity is 25 m/s, and the air tube bulk velocity is 5.3 m/s. These flows yield a mean equivalence ratio of $\bar{\phi} = 0.6$, which is

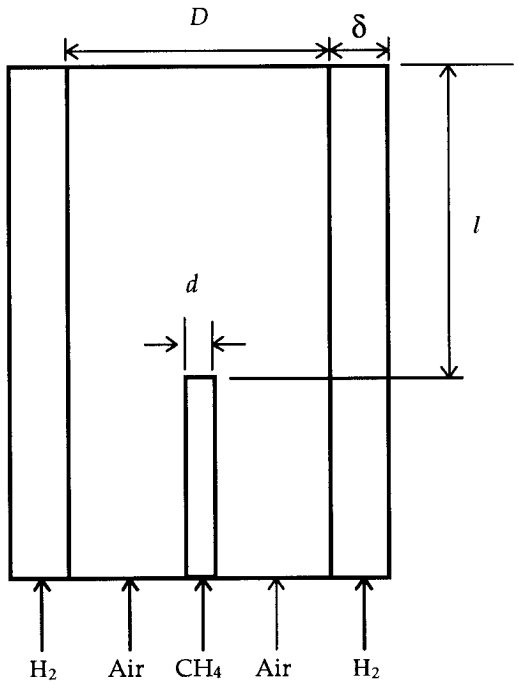

FIG. 2. Schematic of hydrogen-piloted, lean premixed burner. The mixing distance l is varied to change the unmixedness at the burner exit without changing the mean spatial distribution of fuel concentration.

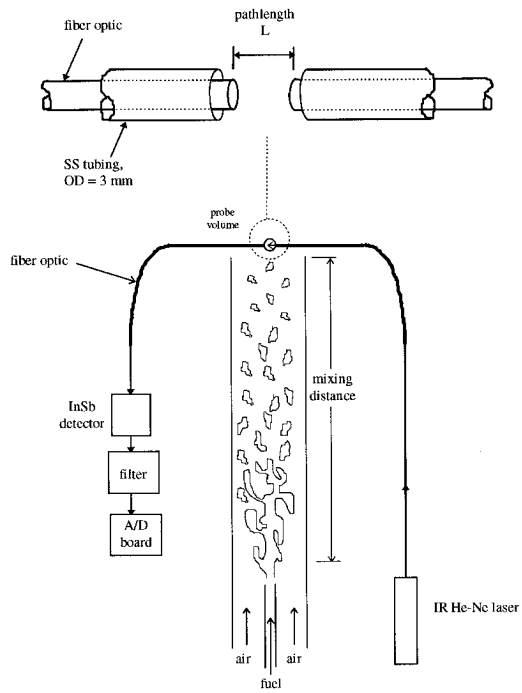

FIG. 3. Schematic of optical probe. A small probe volume is created by the gap between the infrared transmitting fiber optics. Light from a 3.39-μm He-Ne laser is focused into one of the fiber optics.

used consistently for all the test runs.

The radial profile of NO_x with only the hydrogen pilot burning, with co-flow air but no methane flow, confirms that NO_x produced by the hydrogen pilot does not penetrate to the centerline. At a distance of 17.8 cm above the burner exit, the NO_x concentration at the centerline is less than 0.5 ppm. Hermanson and Dimotakis [23] show that increasing the heat release reduces the entrainment of a jet; therefore, the expansion caused by the heat release in the premixed flame would further decrease the entrainment of pilot flame products. Accordingly, we conclude that any NO_x measured at the centerline is produced by the central premixed flame and not by the pilot flame.

Infrared Fiber Optic Probe

A schematic of the optical probe is shown in Fig. 3. Light from a 3.39-μm He-Ne laser is focused into an infrared transmitting fiber optic that guides the laser light to the probe volume. The laser light passes through the probe volume and a portion intersects another infrared transmitting fiber, which guides the light to a liquid nitrogen–cooled indium-antimonide (InSb) detector. The gap width of the probe is L = 5 mm. A low-pass filter with a cutoff frequency of 2 kHz is used before the sampling of the computer data acquisition system that converts the transmission, I/I_o, into a methane concentration. The A/D board samples at 4 kHz, sufficient to resolve fluctuations on the order of 2 kHz (Nyquist criterion). The optical probe sample volume is placed 25 mm above the burner exit along the burner centerline during the nonreacting unmixedness measurements. The probe location is where most of the combustion occurs when ignited.

Planar Laser–Initiated Rayleigh Scattering

Laser-initiated Rayleigh scattering is used to measure the two-dimensional concentration field for the nonreacting flow as described in Ref. 24. A Candela LFDL-20 pulsed dye laser tuned to 590 nm with a pulse duration of 3 μs and pulse energy of approximately 1–2 J is used as the light source. The laser beam, typically 8 mm in diameter, is focused by a cylindrical lens with a focal length of 1000 mm to form a sheet at the measurement location. The center of this laser sheet is located 25 mm above the burner exit. The imaging area is approximately 0.5 mm $\times$ 10 mm $\times$ 60 mm. The images are taken at a repetition rate of 0.5 Hz.

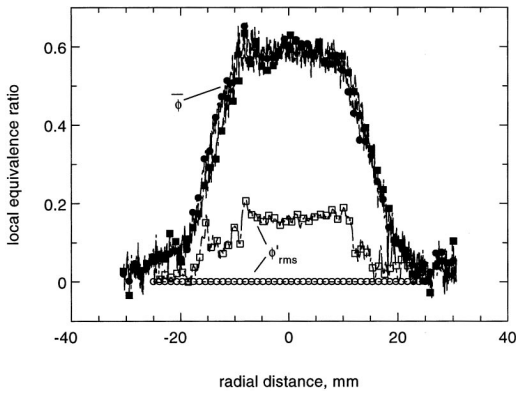

FIG. 4. Mean (solid symbols) and rms (open symbols) of equivalence ratio as a function of radius at an axial distance of 2.54 cm above the burner exit. The circles (both open and solid) are for the perfectly mixed conditions, and the squares (both open and solid) are for the l = 33-cm mixing distance (the most unmixed case studied).

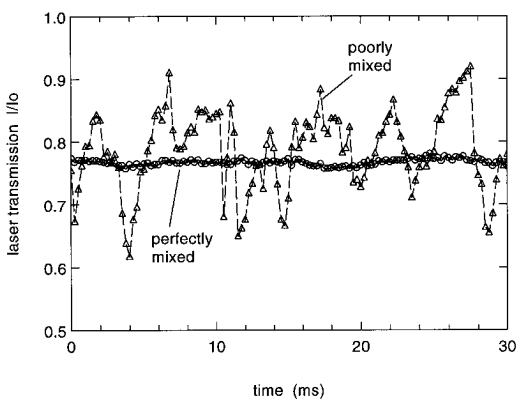

FIG. 5. Laser transmission as a function of time for two extreme mixing conditions. The circles are for the perfectly mixed condition, and the triangles are for the most unmixed condition (mixing distance l = 33 cm). The low values of laser transmission correspond to richer concentrations than the mean, and the high values of laser transmission correspond to leaner concentrations than the mean.

NO_x and O_2 Measurements in Flames

Species measurements are made at an axial location of 17.8 cm above the burner by using a quartz suction probe with a heated sample line as described in a previous work [24]. Horiba analyzers are used for all species measurements. Nitrogen oxides are measured by a chemiluminescent analyzer, CO and CO_2 by NDIR analyzers, and O_2 by a paramagnetic analyzer. Water is removed from the sample upstream of the analyzers by using an ice bath and water trap for the CO, CO_2, and O_2 measurements. The water is removed from the sample entering the NO_x analyzer by using a porous dryer, thus avoiding the problem of NO_2 absorption into water.

Model Description

The effect of unmixedness on NO_x production in the experimental combustion system is modeled numerically using a distribution of equivalence ratios in conjunction with numerical results of a 2-ms PSR in series with a 4-ms PFR. The residence times were chosen for their physical significance (for an exit velocity of 5.5 m/s, a sampling distance of 17.8 cm, and an average temperature of 1600 K; the residence time is approximately 6 ms) and then adjusted to agree with the one experiment. The PSR/PFR code is used to obtain a relationship between NO_x emissions and equivalence ratio assuming the air and fuel are perfectly mixed (i.e., the PSR/PFR code is run several times at different equivalence ratios). The average NO_x emissions of the unmixed combustion system are then computed by

$$NO_{x,avg} = \int NO_x(\phi) \cdot P(\phi) \cdot d\phi \qquad (3)$$

where $NO_x(\phi)$ is the NO_x emissions as a function of equivalence ratio as predicted by the PSR/PFR and $P(\phi)$ is the distribution of equivalence ratios for the combustion system of interest. Consistent with the experiment, $P(\phi)$ has an assumed Gaussian shape with an appropriate mean and unmixedness (or, equivalently, standard deviation).

Results and Discussion

Concentration Measurement in Nonreacting Flows

The mixing distance l was varied between 33.0 cm and 63.5 cm while maintaining a mean equivalence ratio of ϕ = 0.6. A fully premixed experiment was performed as well. The Rayleigh measurements shown in Fig. 4 confirm that the mean equivalence ratio has a uniform profile at the exit of the burner for the mixing distances studied. Figure 4 also shows that as the mixing distance decreases, the rms of the equivalence ratio increases dramatically.

Figure 5 compares the optical probe laser transmission of the fully premixed case ($l = \infty$) to that of the shortest mixing distance (l = 33 cm). Clearly, the large fluctuations in laser transmission for the 33-cm mixing distance indicates large fluctuations in methane concentration. Low levels of laser transmission correspond to high levels of methane concentration (i.e., regions that are richer than the mean). The unmixedness is better quantified by Fig. 6, which shows the distribution of methane concentration for three flow conditions: fully premixed, 57-

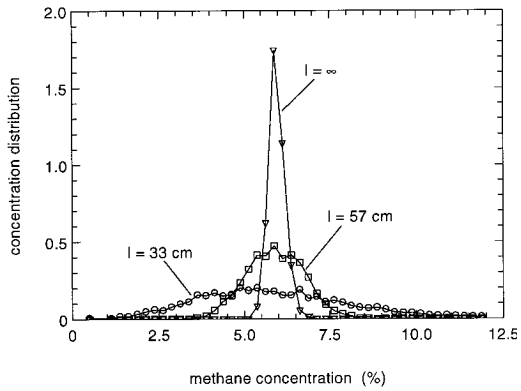

FIG. 6. Distribution of concentration as measured by the optical probe for three mixing conditions. The circles are for the most unmixed case (l = 33 cm); the squares for an intermediate mixing distance (l = 57 cm); and the triangles for the perfectly mixed condition ($l = \infty$). The width of the concentration distribution for the perfectly mixed condition corresponds to the instrument resolution.

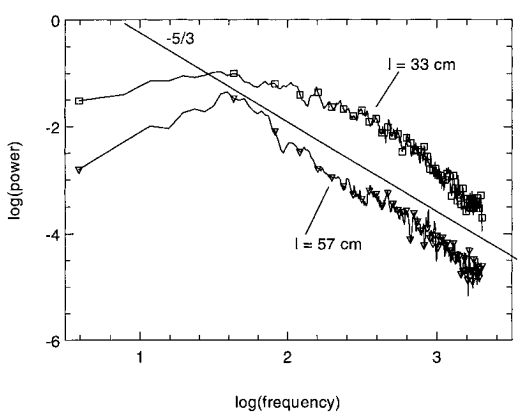

FIG. 7. Scalar power spectrum of concentration for two mixing distances. The squares are for a mixing distance of l = 33 cm, and the triangles are for a mixing distance of l = 57 cm. Both spectra exhibit the expected −5/3 slope cascade; however, the end of the cascade is not observed because the signal was filtered at 2 kHz.

cm mixing distance, and 33-cm mixing distance. The wide distribution of methane concentration for the 33-cm mixing distance includes regions that have near-stoichiometric conditions. Because of the noise in the laser and data acquisition system, the fully premixed flow condition has a measured standard deviation of approximately 0.24% concentration of methane. Therefore, in its current form, the optical probe can measure fluctuations as small as 4% in methane mole fraction (i.e., if $\bar{c} = 0.05$, then $c_{\mathrm{rms,min}} = 0.002$). Improvements to the data acquisition system can be readily implemented.

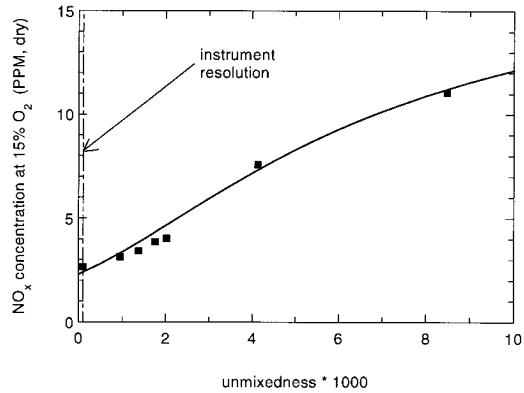

FIG. 8. NO_x emissions (dry) at 15% O_2 versus unmixedness. The symbols correspond to measured NO_x and measured unmixedness (optical probe results). The solid line corresponds to model prediction using a 2-ms PSR in series with a 4-ms PFR in conjunction with a Gaussian equivalence ratio distribution.

The scalar power spectrum of two flow conditions is shown in Fig. 7 as the logarithm of power versus the logarithm of frequency. The cascade of the scalar power spectrum occurs with the expected slope of −5/3 [25]. The end of the cascade associated with a decrease in the slope (more negative) is not observed here because of the present frequency limit of 2 kHz.

Nitrogen Oxide Emissions at Reacting Conditions

The NO_x emissions (dry) corrected to 15% O_2 are shown in Fig. 8 as a function of unmixedness as measured by the optical probe for a mean equivalence ratio of $\bar{\phi} = 0.6$. Also included in Fig. 8 are the modeling predictions of the effect of unmixedness on NO_x formation. As noted previously, for the modeling results, a 2-ms PSR in series with a 4-ms PFR is used to generate the relationship between NO_x and equivalence ratio if the air and fuel are perfectly mixed. At an unmixedness $U = 0.001$, the NO_x emissions are increased by 15% with respect to the NO_x emissions from a fully premixed flame ($U = 0$). At an unmixedness $U = 0.0085$, the NO_x emissions are increased by a factor of 4. The model does an excellent job of predicting the NO_x emissions of the experimental burner, including the slight change of slope that appears at high values of unmixedness. Also shown in Fig. 8 is the resolution of the current optical probe. As depicted by the modeling predictions and the experimental results, the unmixedness resolution $U_{\mathrm{min}} = 0.0001$ is sufficient for measuring whether the air and fuel are mixed well enough for low NO_x emissions.

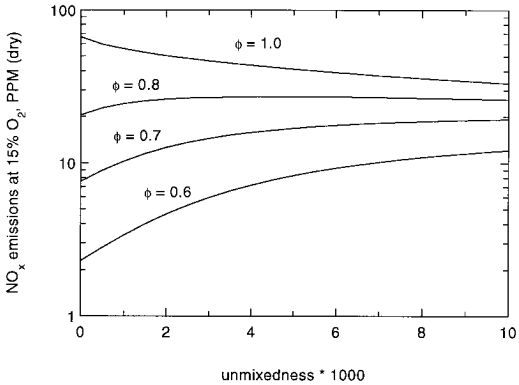

FIG. 9. Effect of unmixedness on average NO_x emissions at different mean equivalence ratios. At lean conditions, the NO_x goes up significantly with increasing unmixedness; however, at a mean equivalence ratio of $\bar{\phi} = 1.0$, increasing unmixedness decreases NO_x emissions.

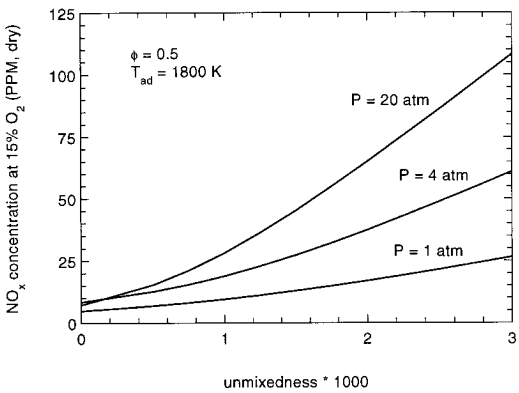

FIG. 10. Effect of unmixedness on average NO_x emissions at different pressures. The NO_x versus equivalence ratio relationship is obtained by using preheated reactants of 700 K and a 2-ms PSR in series with an 18-ms PFR. A mean equivalence ratio of $\bar{\phi} = 0.5$ is used so that the adiabatic flame temperature is 1800 K (typical gas turbine conditions).

Modeling the Effect of Unmixedness on NO_x Emissions

Eq. (3) was used to study the role that the mean equivalence ratio $\bar{\phi}$ plays on the relationship between unmixedness and overall NO_x formation. Graham and McKindley [5] showed experimentally that at *stoichiometric conditions,* the NO_x for poorly premixed combustion was *lower* than the perfectly mixed case; however, at *lean* conditions, the poorly premixed combustion had *higher* NO_x emissions. Figure 9 shows how the unmixedness affects NO_x formation at several equivalence ratios. As shown experimentally by Graham and McKindley [5] and numerically by Fig. 9, the unmixedness effect on NO_x production is significantly different depending on the mean equivalence ratio $\bar{\phi}$.

Modeling the Effect of Unmixedness at Gas Turbine Conditions

Eq. (3) was also extended to gas turbine conditions by using the PSR/PFR code with a 700 K preheat (typical of compressor outlet) to generate the NO_x versus equivalence ratio relationship. The residence time in the PSR was 2 ms, and the residence time in the PFR was 18 ms. Figure 10 shows the effect of unmixedness on NO_x generation at $P = 1$, 4, and 20 atm for a mean equivalence ratio $\bar{\phi} = 0.5$.

The mean equivalence ratio $\bar{\phi} = 0.5$ results in an adiabatic flame temperature of 1800 K. As shown, the unmixedness has a larger influence on NO_x formation at the higher pressures because the peak NO_x levels are larger at the higher pressures. Surprisingly, at low values of unmixedness ($U < 0.0008$), the pressure does not make a large difference in overall NO_x formation. This has some interesting consequences. It has been reported by Leonard and Stegmaier [26] that at sufficiently lean conditions (flame temperature <1900 K), the NO_x from a lean premixed combustor is only a function of flame temperature regardless of the pressure, amount of preheat, or residence time. Figure 10 corroborates the results of Leonard and Stegmaier [26] as long as the unmixedness U is small. For a perfectly mixed combustor ($U = 0$), the model predicts that the NO_x emissions are independent of pressure (proportional to P^0); however, the relationship evolves to being dependent on $P^{1/2}$ as the unmixedness increases.

Conclusions

Lean premixed combustion is an established method of conforming to increasingly restrictive air-quality regulations, especially for NO_x formation. The lower temperatures associated with lean premixed combustion generally lead to lower NO_x emissions; however, the benefit of lean premixed combustion may be lost if the fuel and air are poorly mixed.

A hydrogen-piloted, methane-in-air turbulent flame with a variable premixing distance is used to vary the unmixedness without changing the mean concentration profile. A novel optical probe using absorption of light from an inexpensive 3.39-μm He-Ne laser is used to measure the effectiveness of a premixer. In its current form, the optical probe is capable of measuring unmixedness as low as $U = 0.0001$.

At an overall equivalence ratio of $\phi = 0.6$, unmixedness has a dramatic effect on NO_x emissions.

For an unmixedness of $U = 0.001$, the NO_x emissions are increased by 15% compared to the NO_x emissions of the fully premixed condition ($U = 0$). At an unmixedness of $U = 0.0085$, the NO_x emissions are increased by a factor of 4. The optical probe is capable of resolving the small unmixedness levels that affect NO_x production.

A numerical model of the effect of unmixedness U on NO_x production showed good agreement with experiments. At higher pressures, the model predicts the trend observed by Leonard and Stegmaier [26]. The model shows that at very low levels of unmixedness U, the NO_x emissions are independent of pressure (for an adiabatic flame temperature of 1800 K); however, as the unmixedness increases, pressure plays an increasingly important role in the overall NO_x generated by a lean premixed combustor.

Acknowledgments

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COMMENTS

Giancarlo Benelli, ENEL spa-Thermal Research Centre, Italy. A diagnostic technique that seems interesting for studying a lean premixed combustion system was presented. Did the author estimate the effects of probe design on the monitored mixing process?

Author's Reply. In this paper, we explore the application of the probe to lean premixed combustion. However, we anticipate widespread applications including nonpremixed combustion and piston engines.

The effect of the probe on the mixing process (disturbance of the flow) remains to be quantified. We expect the disturbance is comparable to that associated with the widely used hot wire anemometer.